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Vulnerability Evolution of a Container Shipping Network in an Uncertain Environment: The Case of China–United States Connections

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Abstract: Container transportation has the advantages of standardization, high efficiency, and high safety, which are essential for promoting the development of the world economy and trade. Emergencies such as severe weather, public health incidents, and social security incidents can negatively affect the operational reliability of the container shipping network. To ensure the network security and high-quality operation of container shipping, a double-layer coupled container transportation network is first described to analyze the evolution of the container shipping network and the risk propagation dynamics of operation participants. On this basis, a cascade failure model of the container shipping network considering risk level is constructed. To evaluate the vulnerability of the container shipping network, the transmission mechanism of cascade failure effects of the container shipping network under different emergency development trends and the evolution law and influence path of the container shipping network structure are both analyzed. Finally, we empirically studied the container shipping network in China and the United States, and characteristic parameters of the China–U.S. container shipping network are calculated and analyzed. The model's validity is verified through practical cases and model simulation results, and the cascading failure process of the container shipping network in China and the United States under three types of attacks is simulated. Suggestions are provided for effective improvement in the vulnerability of the container shipping network under every kind of contingency.

Keywords: China–U.S. container shipping network; complex network; vulnerability analysis; cascading failures

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1. Introduction

Maritime transport is the most essential mode of transport in international trade, accounting for more than two-thirds of the total international trade volume, and the container shipping network is a necessary reliance on global economic exchange. The stable production of the port and the orderly operation of the waterway are the conditions required for container transportation to maintain a high level of function. However, the container shipping network is highly dependent on the outside world and is easily affected by bad weather, public health, and social security incidents. The occurrence, diffusion path, and evolution process of the above emergencies are very uncertain, which will break the original balance of container transportation, causing uncertain freight demand and tight capacity of shipping companies, and highlighting the fragility of the container shipping network.

The number of shipping accident emergencies is counted based on the corresponding data of the Maritime Safety Administration of the People's Republic of China, as shown

in Table 1. More than a third of these vessel accidents occurred on container ships. From 2013 to 2021, the annual average number of oil spills in United States waters was 2690, with the highest number of spills in 2013 at 3223 and the lowest in 2021 at 2019 [1]. Moreover, there have been a number of shipping accidents such as wind waves, collisions, and fires in the Pacific Ocean and United States waters in recent years. For example, On 1 December 2020, the vessel ONE Apus of ONE was sailing from China to the United States when it encountered winds and large swells near Hawaii in the North Pacific Ocean, it is estimated that the number of lost or damaged containers exceeded 1900 [2]. On 8 July 2023, the barge touched the dock when it docked at Lake Charles Harbor, the damage to the dock was approximately USD 1 million. On 17 February 2023, while passing through Lake Salvador in Perot Bay, Louisiana, a fire broke out in the tugboat Desperado's engine room, resulting in a total of USD 30,000 in damage to the vessel [3].

Table 1. Statistics on the number of shipping accidents in China by type in the last ten years.

	2013–2016	2017–2020	2021–Now	Total
Collision accidents	119	126	70	315
Stranding accident	7	8	3	18
Accidents on the rocks	7	20	1	28
Touch accidents	14	23	12	49
Fire/explosion accidents	16	16	2	34
Wind accidents	10	20	2	32
Self-sinking accident	65	57	13	135
Other accidents	64	85	42	191

In addition, demand fluctuation and capacity constraints led to severe port congestion and supply chain disruption due to the unexpected COVID-19 pandemic in early 2020. According to the China Export Container Freight Index released by the Shanghai Shipping Exchange, the change-over-time relationship between supply and demand in the container market and freight is shown in Figure 1. In the fourth quarter of 2019, freight rates increased slightly due to increased demand before the Chinese New Year holiday. In the first quarter of 2020, the COVID-19 outbreak caused economic and production activity to stall, resulting in lower demand, major shipping companies stopped sailing, and freight rates fell. In the second quarter of 2020, the spread of COVID-19 in Europe and the United States led to a continuous decline in demand, shipping companies took measures such as stopping flights to maintain the balance of supply and demand, and freight rates continued to fall. In July 2020, freight rates rebounded due to the recovery of demand driven by countries' initiatives to stimulate economic recovery and the level of capacity disruptions remained high. In September 2020, shipping demand increased as the domestic epidemic situation improved and coincided with the peak shipping season. In December 2020, the total export volume in China's daily necessities industry increased by 20.5% year-on-year, including epidemic prevention supplies and daily necessities. These sharp increases in demand intensified container demand. However, freight rates rapidly increased due to the low turnover rate of container ships and containers and the continuous decline in liner punctuality, resulting in the spread of global capacity imbalance. In the first quarter of 2021, the shipping demand continued to grow, benefiting from China's efficient epidemic prevention and control measures; however, the supply of spaces was insufficient, and freight rates soared.

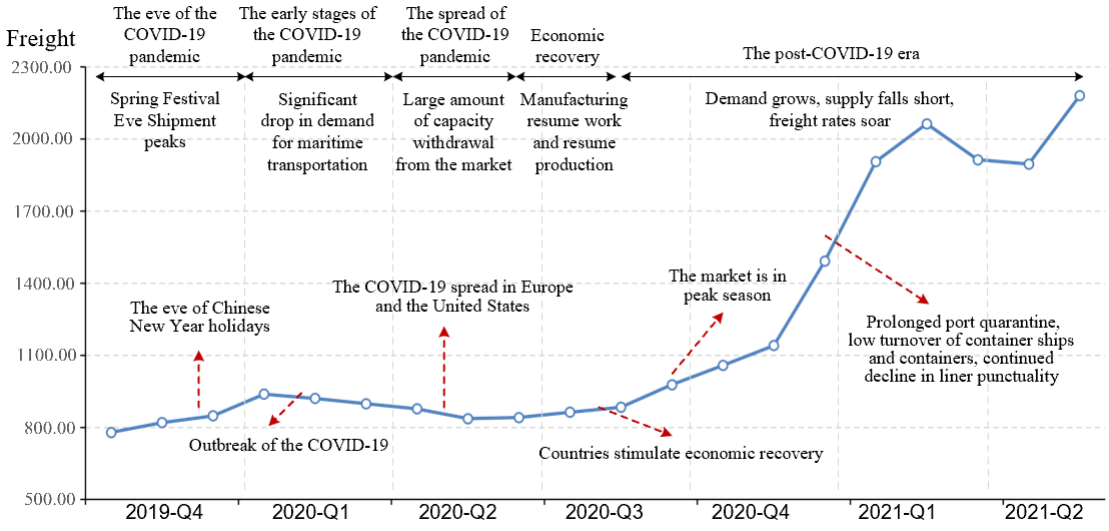


Figure 1. The route freight index chart from 2019-Q4 to 2021-Q2.

Given the above situation, a double-layer coupled container shipping network is first constructed, which includes the whole process from distribution to port operation. Then, a cascade failure model of the container shipping network considering the risk level is constructed to depict the impact of the transmission mechanism of emergencies on the container shipping network. Finally, diversified scenarios are designed according to the risk level of emergencies, and vulnerabilities of the container shipping network under different scenarios are assessed.

For the container shipping industry, the complicated shipping process and various emergency categories bring difficulties in analyzing the impacts of the emergency transmission mechanism on the container shipping network. The contributions of this paper are primarily demonstrated in three dimensions.

1. Based on the integration of susceptible infected recovered model and complex network theory, an analytical framework is proposed for risk propagation in a double-layer coupled container shipping network. Considering the risk levels, a cascade failure model is constructed for quantifying the transmission mechanism of emergencies through the container shipping network.
2. Taking network hierarchy and emergency category into consideration, a simulation procedure is organized for visualizing the cascade failure spreading along the container shipping network. Vulnerability assessment criteria are formulated to quantify the extent of influence, including average failure scale and average network efficiency.
3. Practically, China–United States container shipping network is selected to verify the effectiveness of the proposed model. Diversified scenarios are designed according to the risk level of emergencies. Three events (public health event, Suez Canal event, and strike event) are simulated to elucidate the influence paths and describe the evolution of the container shipping network. Meanwhile, suggestions are provided to improve the damage resistance of the shipping networks.

2. Literature Review

Our research is mainly related to the container shipping network from the perspective of complex networks, risk identification and emergency assessment, and vulnerability in the field of shipping networks.

2.1. Complexity of Container Shipping Network

The global container shipping network is one of the world's most complex networks, and it has been extensively studied. Ducruet [4] focused on analyzing shipping networks in different regions and concluded that shipping networks in different areas vary significantly. Ducruet et al. [5] verified the correlation between network characteristics and container throughput. By comparing the centrality of ports in the Atlantic Ocean between 1996 and 2006, the structural scale of the maritime shipping network has increased significantly, but most of the global traffic flow is still limited to short distances. Considering the scale-free and small-world characteristics, the complex network model could more intuitively show the ports' position in global transportation systems. Bian et al. [6] changed the number of associated ports and correlation intensity of the shipping network and found that the centrality of port nodes can be calculated by the number of connected nodes and the average weight of these nodes. Wang et al. [7] studied the centrality degree of the shipping network and evaluated the influence of port operation capacity and hinterland market economic coverage. The results showed that the hinterland market coverage significantly affected the demand and closeness of ports. After studying the ports of China, Russia, South Korea, and other countries, Ducruet et al. [8] proved that the evolution of shipping network design was significantly affected by the policies of port cities. Dirzka et al. [9] and Lim et al. [10] used experienced capacity data from container shipping companies to analyze ports and routes embedded in the shipping chain. The results showed that changes in geography and the composition of participants might impact the dynamics of port-oriented shipping supply chains. Li et al. [11] studied the network service capability of container shipping networks and proposed a service capability reconstruction coefficient incorporating the port location coefficient, inter-port distance importance coefficient, and inter-port route importance coefficient. The case study of an existing container shipping network revealed that little adjustments are required to enhance its balance and stability. Qin et al. [12] innovatively constructed a three-dimensional econometric model of port resilience based on port nodes' linkages and port attributes in the shipping network. Guo et al. [13] proposed a coordinated mechanism for global synchromodal transport planning to enhance the total profits, taking into account many operators across diverse routes within a complex shipping network.

2.2. Risk Identification and Emergency Assessment

Risk identification is the recognition of actual or potential risks by perception, judgment, or categorization, serving as the initial stage and foundation of risk management and control. Risk assessment denoted the quantitative evaluation of the potential effects and likelihood of losses affecting people's lives, property, and other aspects before or during an occurrence[14,15]. It is challenging to achieve risk identification in container shipping, owing to the inherent fragmentation and overlap in multiple container shipping operations [16]. Initially, the risk level of emergencies was assessed by surveys, expert interviews, and assessments of ports by specialists. Recently, the identification and assessment of risk in the shipping industry have shifted from qualitative to quantitative methods. Quantitative evaluation methods have predominantly been adopted, including systems engineering theory, pattern recognition, artificial intelligence, and other theories for identification and rating assessments [17,18]. Huntington et al. [19] developed a risk avoidance model by examining factors, such as speed and collision avoidance zones, and explored the relationship between these factors and navigation hazards. Huntington et al. [20] examined the danger of an oil leak resulting from a marine collision and built a collision risk model for the maritime transport system in an unpredictable environment. Kabir et al. [21] established a risk assessment framework following an evaluation of the collision risk associated with ro-ro passenger ships at sea. In the case study, they analyzed the potential risks of maritime transport utilizing the Gulf of Finland as the entry point. Based on Bayes-

ian statistics, Sanchez-Gonzalez et al. [22] analyzed the accident data from five ports, including variables such as visibility, tide, and channel complexity. The quantitative outcomes of ship stranding probability were also captured. Baştuğ et al. [23] reviewed the marketing risks in the shipping industry and emphasized the principal marketing risks and corresponding risk mitigation strategies to guarantee resilience during crises.

2.3. System Vulnerability

Vulnerability is often defined as the sensitive attribute of the system when the system is attacked. More and more scholars have begun to focus on the concept of vulnerability and engage in its study, even on network vulnerability. Ghazavi and Ebrahimi [24] performed an investigation of groundwater vulnerability and found that it typically represented the self-protective response against external pollution. The majority of research on vulnerability within maritime networks has been conducted from both global and regional perspectives. Bešinović [25] elucidated the current situation and developmental trends of the road transport system, offering a reference for vulnerability research in transportation sections including railroads, aviation, and shipping. Based on analyzing the vulnerabilities of the global container shipping network, Peng et al. [26] put forward feasible recommendations for port construction and optimization of port operations. Feyrer [27] utilized graph theory to practically analyze the global container shipping network. Alongside examining alterations in the comprehensive structure of the network throughout the long term (from the 1970s to the present) and the mid-term (1970s), they also investigated modifications in the container shipping network before and after the reopening of the Suez Canal. Viljoen and Joubert [28] implemented and assessed two link-based interrupt strategies on a global container shipping network to simulate the impacts of large-scale service reconfiguration on priority links. The findings indicated that both strategies would reduce network flexibility, leading to a decrease in transshipment and dynamic rerouting capabilities among the most congested port areas. Xu et al. [29] analyzed the impacts on vulnerability, reliability, and potential risk of the global container shipping network when a port was interrupted. Jiang et al. [30] selected the 21st-century Maritime Silk Road container shipping network for topology analysis, and the vulnerability of the maritime network to cascade failure was examined according to the route connectivity and the empirical port load data. Yue and Mangan [31] provided a three-dimensional framework for analyzing the reliability in container shipping networks, focusing on infrastructure reliability, network configuration reliability, and connectivity reliability.

The current methodologies and findings in three domains, complex network, risk identification, and system vulnerability, have established a foundation for assessing the vulnerability of shipping networks. This paper studies the container shipping network, which possesses distinctive characteristics that have been relatively unexplored in the current literature. Firstly, the container shipping process is executed by integrating the mainline shipping with the feeder shipping. The mainline network for ocean shipping and the feeder network for collection and distribution are interconnected and intricate. Such a connected network is distinctly dissimilar to the conventional complex network topology. The construction of container shipping networks is fundamental for studying network vulnerability, but there is limited related research. Secondly, the container shipping network has many port nodes with various hierarchies. Within a shipping network, there exist links among port nodes of identical levels as well as nodes of different levels. Considering disrupting the operational environment of container shipping, emergencies might lead to multi-stage risk propagation, hence impacting network connection and structure. Traditional methodologies of risk identification and risk assessment inadequately explain the cascading impact process. Thirdly, the whole container shipping process involves multiple steps causing long shipping times, and various types of emergencies may occur. In response to different emergencies, there are distinct disparities in the

resilience and vulnerability of container shipping networks that require thorough evaluation. In this paper, the vulnerability of shipping networks in uncertain environments is studied considering the comprehensive shipping network and diversity of emergencies.

3. Methodology and Data

3.1. Problem Description

The logistics system of the container shipping network is an open and information-sharing platform. Container transportation refers to loading goods into containers, using containers as cargo collection units, loading and unloading and transportation operations, and realizing the transportation of goods from "door to door". The construction of container shipping network topology is the basis for studying the characteristics of the shipping network. The port is abstracted as the network node, and the route is outlined as the edge of the network by mapping into a topological diagram in complex network theory. Through mapping into topological graphs in complex network theory, this paper abstracts ports as nodes of the network and routes as the edges of the network. In this paper, the L-space model is chosen as the representation method for constructing the container shipping network, in which a node represents each port, and the connection between any pair of adjacent ports on one or more lines is defined by an edge.

When goods traverse in the shipping network, they must be transshipped through the inland river collection and distribution network, coupling the shipping and technical cloud network. In the event of congestion risk in the shipping network, the port assumes a pivotal role as the source of infection for the spread of congestion risk. If the ports in the shipping network are unable to absorb the shock, the burden will be transmitted to the corresponding port nodes of the collection and distribution network, leading to the spread of risk in the collection and distribution network. In this context, the maritime network is defined as the base network.

In the coupled network, there are two types of networks: the base network layer and the external network layer [32]. Each network layer has an interconnected relationship among nodes in the same layer. There is also a link relationship and coupling effect between nodes in different network layers. Unlike a multi-tier network, the coupling system base network connects the two sub-networks through port nodes. Congestion risk propagation is transmitted laterally within the underlying network and between the basal and external network layers. Figure 2 is a schematic diagram of the coupled network system based on the available relevant research.

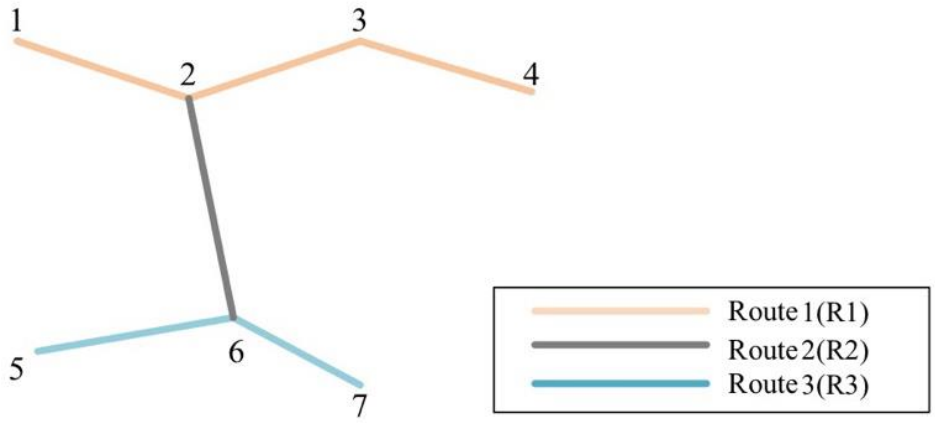


Figure 2. Schematic diagram of a simple container shipping network.

Figure 3 shows a schematic diagram of the coupled container shipping network, where there are two sub-networks, the maritime network (A) and the collection and distribution network (B). Nodes a and c in the maritime network correspond to different seaports, while the edges connecting two nodes denote legs of shipping routes. In the collection and distribution network, nodes b and d correspond to different inland ports or dry ports, and the edges represent the extension of shipping routes to inland areas. The coupled container shipping network is integrated by connecting the two sub-networks through port nodes, where they are functionally complementary. The congestion risk propagation is not only transmitted horizontally within a sub-network but also transmitted from the maritime network to the collection and distribution network. When a risk occurs at node a, it will be propagated within the maritime sub-network and gradually transmitted to node c. Sub-network B is affected by sub-network A, node b in sub-network B obtains information from the corresponding coupled node a in sub-network A and propagates this information within the sub-network B. Through the propagation of the inner neighbor, node d appears, which is affected by the propagation of operational risks.

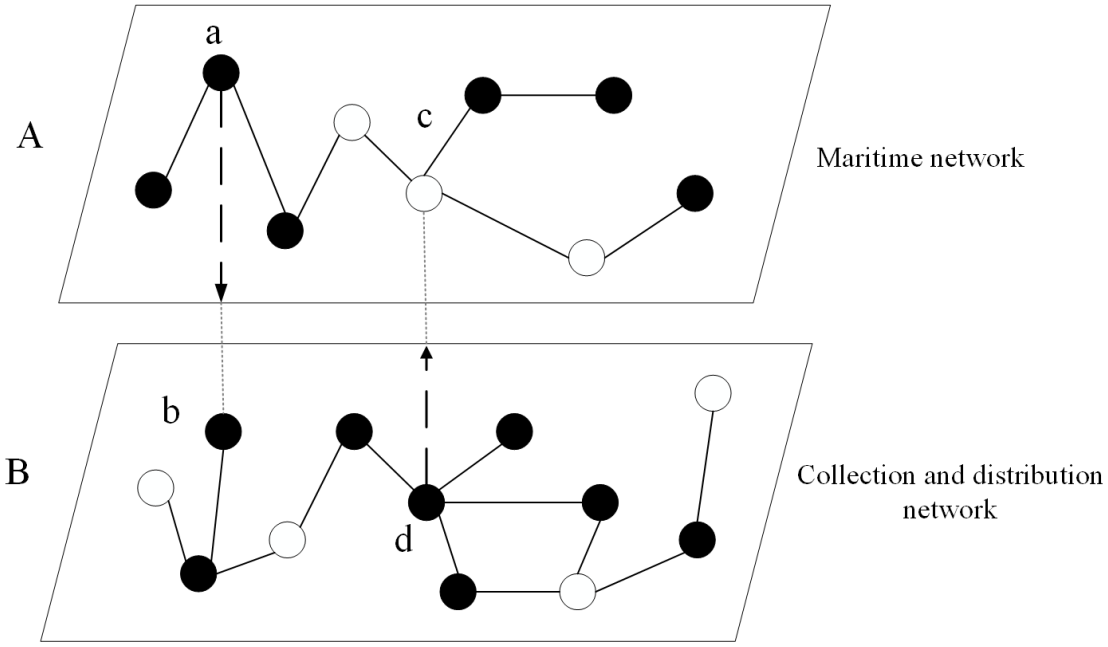


Figure 3. The multi-layer coupled container shipping network.

According to the characteristics of the container shipping network, the following assumptions are made on the topology model of the container shipping network:

(1) There are multiple terminals in one port. Although these terminals belong to different operators, they are nevertheless close to each other. We disregard the notion of the terminal and consider the ports as the pivotal nodes of the China–U.S. container shipping network.

(2) The direction of ship navigation is bi-directional, but we only consider the undirected network structure. Here, the China–U.S. container shipping network is abstracted as an undirected network.

(3) In the actual operation process, China–U.S. containerized maritime transportation has multiple transportation routes between two cities. When building the China–U.S. container shipping network, since there are no two edges with the same starting and ending points in the network, only one edge is realistic.

(4) The container shipping network between China and the United States is a multi-layered system, consisting of the shipping network and the crucial collection and distribution network. Risk to the shipping network layer can be passed to the collection and distribution network layer. However, the throughput in the collection and distribution

network is not of the same magnitude as that in the shipping network, and it is not easy to impact the shipping network.

(5) The routes of the container shipping network in China and the United States are different from some commercial route data. The routes in this paper include a departure port, a destination port, and an unlimited number of transit ports. A flight segment refers to a pair of call ports, with only the departure port and the destination port.

3.2. Cascade Failure Model for the Container Shipping Network

In a complex network, the failure of one or a few nodes or connections will cause other nodes to fail through the coupling relationship in the container shipping network, resulting in a cascade effect, eventually leading to the collapse of a considerable number of nodes or even the entire network, this phenomenon is called cascade failure. According to the characteristics of the container shipping network, a cascade failure model based on the multi-layer coupled container shipping network is constructed. The failure process of the container shipping network is described from a macro perspective; that is, the mutual constraint relationship between container freight volume and ships is not considered. In this paper, the network will not be resilient once the port node fails.

The port load is generally expressed as the total weight or cargo volume that is loaded onto or unloaded from all of the container ships at a port during a specific period. This measurement could reflect the operation volume and throughput of the port. Therefore, in the calculation of port load, this paper describes the loading and unloading operation process of containers considering the temporary storage, introduction, and transit operation of containers in ports. The function of port load is defined as follows:

$$L_i(0) = O_i + D_i - B_i \tag{1}$$

In Equation (1), $L_i(0)$ is the initial load of the port, O_i is the number of containers temporarily stored in port i in the initial state, that is, the initial stock of port i containers. D_i is the volume of containers that is transferred into port i, and B_i is the amount of containers that is transferred out of port i. When the container shipping network is operating normally, each port has a corresponding initial load, and the port load will also change when the container shipping network has an operational risk shock and cascading failure effect.

Port capacity represents the maximum amount of cargo that a port or terminal can handle within a given time, often expressed in terms of TEUs (twenty-foot equivalent units) for containers. Port capacity reflects the design specifications, available infrastructure, and resources, and is crucial for planning and operational management. In the container shipping network, ships travel along established routes. Since port nodes are affected by the size, area, length of berths, and carrying capacity, the container operation volume of ports and related roads is limited. Based on the complex network theory and the Motter–Lai model, the collection and distribution capacity of the port node in the topology network model of a container shipping network can be described as the capacity of the port node to satisfy the operation of the container network and prevent the phenomenon of cascading failure. According to the Motter–Lai model, the security capacity of port nodes can be obtained as follows:

$$C_i = (1 + \alpha_1) L_i(0) \tag{2}$$

In Equation (2), C_i is the safe capacity of the port i. Safe capacity is the level of cargo handling that can be managed without compromising safety regulations or operational efficiency. This capacity takes into account factors such as equipment limitations, labor availability, and environmental conditions to ensure safe and effective operations. $L_i(0)$ is the initial capacity of the port i, reflecting the baseline level of cargo handling

capability of a port or terminal at the outset of its operations. Here, we set the initial capacity as port load. α_1 is the tolerance coefficient of the port node, which represents the redundant capacity of the port node for container load, and $\alpha_1 > 0$.

When the shipping network operates, the port's operation volume exceeds the safe capacity of the port, resulting in a decrease in the port's operational efficiency and an increased risk of port security accidents. At this point, container operations within the port system can remain unhealthy. When the port operation volume surges, or for other reasons leads to the interruption of the process, the port node state is defined as the failure state. This paper establishes the port node state as the failure state, and the relationship between the limit state and the initial capacity according to the Motter–Lai model is as follows:

$$S_i = (1+\alpha_2) L_i(0) \tag{3}$$

In Equation (3), S_i is the limit capacity of port i, α_2 is the limit coefficient of port i. Limit capacity represents the maximum threshold of cargo that a port or terminal can manage before facing significant operational risks or failures. Exceeding this limit can lead to congestion, delays, and potential safety hazards, thereby affecting the overall performance of the port. The greater the tolerance and limit coefficients of a container shipping network, the greater the ability of the network to cope with cascading failure effects. But in fact, the larger the container throughput, the higher the cost, so α cannot be increased indefinitely.

According to previous Motter–Lai model studies, most analyses assume that there are only two types of port node states during cascading failures, which is normal and fail [33]. In these studies, port nodes are removed from the network when they exceed their safe load. However, in the actual network, when the port load exceeds its safe capacity within a specific range, the port can operate generally inefficiently with a particular risk of failure. Therefore, when the port's operation volume appropriately exceeds its capacity for the container shipping network, the port node is partially successful. The operational efficiency of the port node and the service level of the port both decrease, the container turnover is limited, and operational risks spread in the port network. In this case, it is possible to restore the port node into a normal state without any impact on the network by the timely emergency treatment. If the port's operational volume continues to increase, and the port cannot continue operations when it grows to a certain extent, then the network node is in a failed state. Based on the above analysis, considering the actual procedure of port nodes, this paper divides the port node status in the cascade failure process of the container shipping network into normal port nodes, overloaded port nodes, and failed port nodes, which are defined as follows:

$$Li(t+1) = \begin{cases} 0, 0 < Li(t) \le Ci \\ Li(t) - Li(0), Ci < Li(t) \le Si \\ Li(t), Li(t) > Si \end{cases} \tag{4}$$

In Equation (4), $Li(t+1)$ indicates the load amount that the port i distributes to its neighboring port nodes at the next step after the network is attacked, and $Li(t)$ indicates the port load of port i at step t.

When $0 < Li(t) \le Ci(0)$, the load of port i is less than the port capacity, then the port node is in a normal state. A normal state means the port node is in normal condition, and there is no need to distribute the port load to the adjacent port nodes at the next moment. When $Ci(0) < Li(t) \le Si(0)$, the total cargo volume is larger than its initial capacity but less than its maximum capacity, then the port node is in an overloaded state. In the overloaded state, the container port could operate normally, but the efficiency would be affected. When $Li(t) > Si(0)$, the total container cargoes needed to be handled is much larger than limit capacity of the port, and the port node is in a failed state. In the next moment, the total load of the failed port would be distributed to its neighboring port nodes.

The abnormal occurrence of a node in the China–U.S. container shipping network, such as capacity reduction or paralysis, will have impacts on other related nodes in the maritime network and the collection and distribution network, even having an impact on the operation efficiency of the entire network. The process of such affection is called risk diffusion. The risk diffusion process in the container shipping network between China and the United States has heterogeneity and complexity. Based on the Motter–Lai model, this paper introduces the concept of infection probability in the SIR model, which is a classical model in virus propagation theory. The infection probability describes the state change of nodes during the cascade failure of a container shipping network. The SIR model mainly considers three states of individual existence, which are susceptibility state (S), infection state (I), and removal state (R). In this paper, the container shipping network corresponds to the SIR model, the standard port node corresponds to the susceptible node in the SIR model, the overloaded port node corresponds to the infection node, and the failed port node corresponds to the removal node. When the port is in a normal state, if the adjacent port node is in an overload state, it will be spread by the overloaded port node with λ as the node in the overload state, where λ represents the probability of port operation risk propagation. Meanwhile, if the congested port node continues to increase the load and exceed its limit capacity, the node status will change to a failed node. A node port in a failed state does not propagate the risk to other port nodes. The change process of port node status based on the cascade failure model of a multi-layer coupled container shipping network is shown in Figure 4.

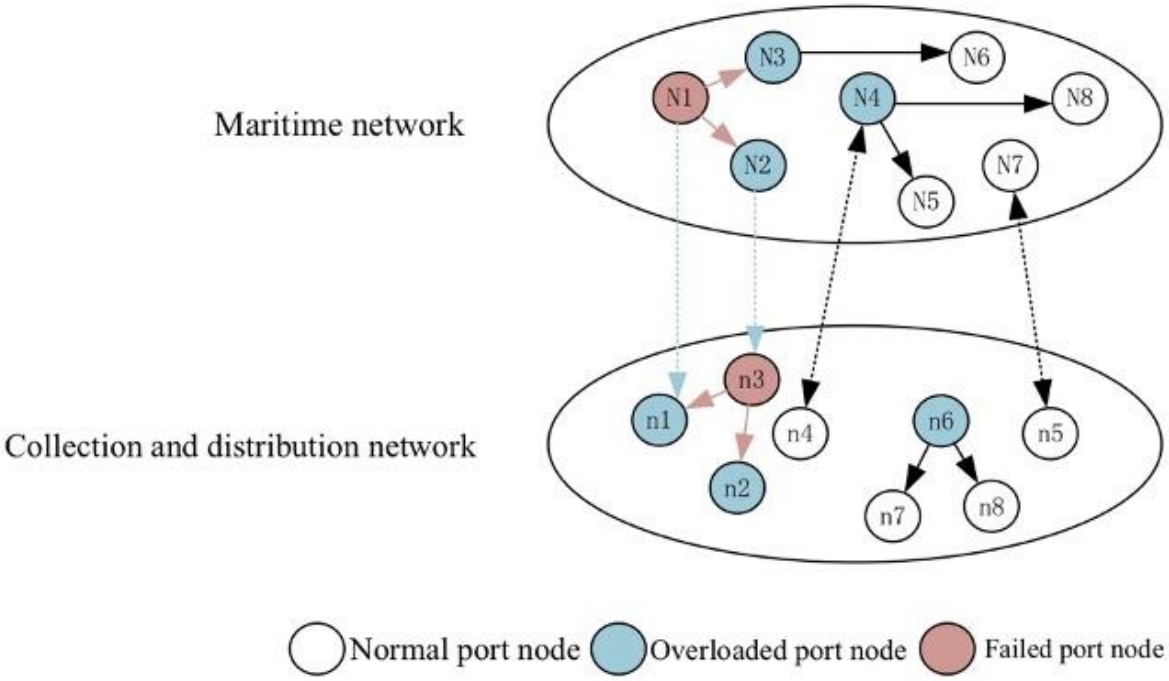


Figure 4. The change process of port node state in a cascading failure model for the container shipping network based on node state change.

$s(t)$, $i(t)$, and $r(t)$ represent the normal port nodes, overloaded port nodes, and failed port nodes in the time t, respectively. The dynamic equation of the cascade failure model of the container shipping network based on the change in node state can be expressed by the differential Equation (5).

$$\begin{cases} \dfrac{ds(t)}{dt} = -\lambda i(t)s(t) \\ \dfrac{di(t)}{dt} = \lambda i(t)s(t) - \mu i(t) \\ \dfrac{dr(t)}{dt} = \mu i(t) \end{cases} \tag{5}$$

In Equation (5), μ represents the probability of the port from overload state to failure state. The most critical indicator parameter in operational risk propagation in a double-layer coupled container shipping network is the propagation threshold [34]. The propagation threshold refers to the port node in the adjacency state of overload and failure. Whether the operational risk can complete the diffusion process in the container shipping network depends on whether the port operational risk propagation probability λ is greater than the propagation threshold λ_c.

The comprehensive service level factor of the port mainly depends on the punctuality rate and the density of calls. Whether the ship can arrive at the port on time is an essential indicator of the service level of liner shipping companies and port companies and a critical indicator of liner shipping and operating costs. According to the different punctuality rates of the port, the comprehensive service level of the port is quantified, as shown in Equations (6) and (7).

$$Ser\left(u_i, u_j\right) = SR \times \rho_{PC} \times 100 \tag{6}$$

$$\rho_{PC} = \frac{N_{PC}}{S_{PC}} \tag{7}$$

In Equations (6) and (7), Ser is the comprehensive service level of the port, SR is the on-time shift rate, ρ_{PC} is the call density, N_{PC} is the number of calls at the port, and S_{PC} is the total number of port calls.

The associated congestion costs are quantified by studying port flow and vessel delay time under different parameters. Studies have shown a significant exponential relationship between vessel delays in port and port cargo volumes [35,36]. The time of the vessel in port is quantified as follows:

$$tim\left(u_i\right) = a\left(u_i\right)^b \tag{8}$$

In Equation (8), $tim\left(u_i\right)$ is the delay time of the ship waiting in port, u_i is the throughput of port i, a is a constant, which satisfies $a > 0$, and b is a constant, which satisfies $b \geq 1$.

The topological network factor mainly depends on the modality of the shipping network. Nodality is the number of edges associated with port nodes, as shown in Equation (9).

$$a_{ij} = \begin{cases} 1, u_i u_j \in E\left(G\right) \\ 0, u_i u_j \notin E\left(G\right) \end{cases} i, j = 1, 2, 3, \ldots, N \tag{9}$$

In a container shipping network, the modality can be expressed as Equation (10):

$$Deg\left(u_i\right) = \sum_{j=1}^{N} a_{ij} \tag{10}$$

In this paper, λ denotes the probability that a normal port node will be affected by an overloaded port, which is the probability of operational risk diffusion. There is a tight coupling between the various port facilities in the container shipping network. The tight

coupling makes the entire container shipping network react quickly to any disturbance, and correspondingly, the trouble will spread rapidly in the container shipping network, affecting the regular operation of the entire container shipping network.

With the continuous development of the container shipping network, the density of the container shipping network has increased significantly, and the correlation and dependence between ports and ports, harbors and routes, and routes and routes have become stronger and stronger. This increases the operational risk of overloading port nodes. The overloaded port node is similar to the infectious disease model's infection status (I) individuals. Congested port nodes affect the operational levels of their adjacent normal nodes, while congested port nodes themselves may fail and be removed from the container shipping network. The probability of operational risk diffusion is described in Equation (11).

$$\lambda = K_1\theta_1 + K_2\theta_2 + K_3\theta_3 \tag{11}$$

In Equation (11), K_1, K_2, and K_3 represent the weight factor corresponding to each element, respectively, and $K_1 + K_2 + K_3 \leq 1$.

Among the factors influencing the comprehensive service level of ports, the establishment of the topology network is affected by whether there are flights. When there is a flow of goods between port A and port B, then port A and port B establish contact. The impact of the overall integrated service level could be determined based on the cargo volume between ports A and B and the cargo situation of each port, which is θ_1.

Among the factors influencing the berthing situation of ships in port, the establishment of a topological network is affected by the waiting time of ships in port. When the vessel waits at port A for a certain amount of time, port A of origin establishes contact with destination B. By the similarity of the attribute influencing factors, the influence of the port congestion influencing factors can be determined, which is θ_2.

In the network topology, the establishment of the topological network is affected by the number of port calls. When the frequency of port traffic reaches a threshold in the entire shipping network, ports are linked. The impact of overall topological can be determined by analyzing the number of port calls, which is θ_3.

When the container operation volume of the port is less than the safe capacity of the port node, it is in a normal state, and the port node can continue to operate normally in the container shipping network. When an unexpected event occurs at a port node or its operational capacity exceeds its limit capacity, the state of the port node transitions to a failed state, and the port node will not be able to operate normally in the container shipping network. A specific redistribution method will be used to redistribute the container operation volume of the port node. When the port node is in an overloaded state, the operation volume of the port node is greater than the safe capacity of the port node and lower than its limit capacity, and the node is in a fragile state, which is quickly interrupted and removed from the network due to equipment failure or worker strike.

Considering the redundancy of port nodes and the schedulability of additional loads, this paper introduces the concept of failure probability into the cascade failure model of the container shipping network. That is, when the throughput of a port node is less than its safe capacity, the port node does not fail. When the throughput of a port node exceeds its safe capacity but is less than the limit capacity, the port node fails with a certain probability. When the throughput of a port node exceeds its limit capacity, the port node immediately fails. In this paper, p_i refers to the failure probability of port nodes, and it is assumed that the failure probability of interlayer transmission of overloaded port nodes follows a uniform distribution, as shown in Figure 5.

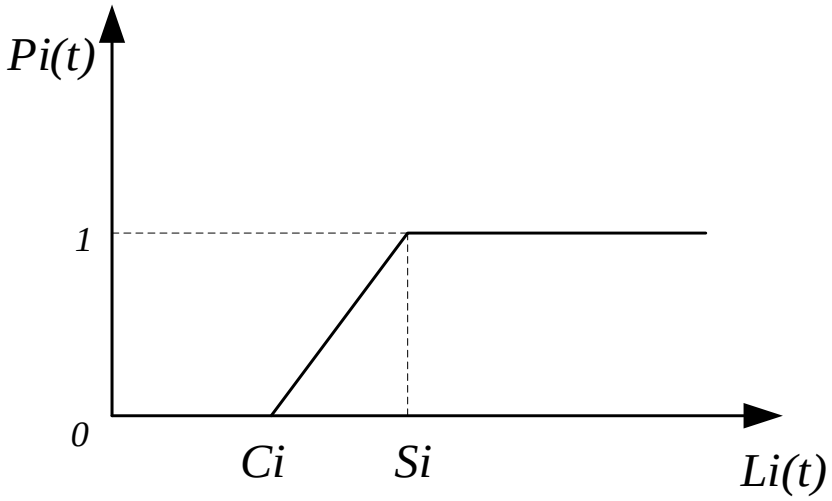


Figure 5. Overload node interlayer risk propagation failure probability distribution.

The failure probability of interlayer risk propagation is expressed by Equation (12).

$$p_i(t)=\begin{cases}0, 0<Li(t)\le Ci(0)\\ \dfrac{Li(t)-Li(0)}{Ci(0)-Li(0)}, Ci(0)<Li(t)\le Si(0)\\ 1, Si(t)>Li(t)\end{cases} \tag{12}$$

To better analyze the dynamic vulnerability and cascading propagation of the container shipping network in China and the United States, the following principle is followed, whereby if a city has both seaports and inland ports, the interaction between the network composed of the two ports can be expressed through the interlayer link. Then, the topological characteristics and dynamic vulnerability of the network can be analyzed.

Since the flow of goods is bidirectional, and the flow of goods on different lines is unequal, the container shipping network in this paper is an undirected weighted network. The cargo can be transported along the network structure in two directions. At each level, ports are coupled between transport routes and cargo flows. Considering that port nodes are affected by environmental factors and additive effects, ports will become increasingly vulnerable after being exposed to multiple risks of collapse. However, ports have a specific self-healing ability, which will not cause the port to collapse within the safe load range and only when the safe load is exceeded and the limit load is not reached. If the safe load is exceeded, it crashes completely. Interlayer transfer is expressed by Equation (13).

$$L_m(t+1)=L_m(t)+p_m L_{m'}(t) \tag{13}$$

In Equation (13), $L_{m'}(t)$ represents the load of the port node m in the shipping network corresponding to the port node m' in the collection and distribution network at time t.

When a port node i in the container shipping network is in a failed state, it is necessary to take a specific load distribution method to distribute the port's operation volume to its neighboring ports. A variety of load redistribution methods have been proposed in existing studies. In this paper, the residual capacity allocation method [37,38] is selected for load redistribution. This method considers the proportion of the remaining capacity of adjacent port nodes for allocation, which is more practical. When the port node is overloaded, the next moment load of the port node j adjacent to the port node i is expressed as Equation (14).

$$L_j(t+1) = L_j(t) + \lambda(1-p_i)\left(L_i(t) - C_i(0)\right)\frac{C_j(t) - L_j(t)}{\sum(C_{ki}(t) - L_{ki}(t))} \tag{14}$$

In Equation (14), $\sum(C_{ki}(t) - L_{ki}(t))$ is the sum of container vacancies adjacent to i.

When a node fails, if node i fails at time t, then the load of the next moment of node j adjacent to node i is expressed as Equation (15).

$$L_j(t+1) = L_j(t) + \frac{\lambda(1-p_i)\left(C_j(t) - L_j(t)\right)}{\sum(C_{ki}(t) - L_{ki}(t))} \tag{15}$$

3.3. Simulation Procedure of Cascade Failure Spreading along the Container Shipping Network

The vulnerability assessment index of the container shipping network is a crucial issue for assessing and controlling the vulnerability of the container shipping network. The average failure scale and average network efficiency are selected to evaluate the exposure of the double-layer coupled container shipping network.

The average failure scale S is a natural indicator of the degree of cascading failures, and the ratio of the number of failed port nodes to the total number of nodes is the average failure scale for the entire network. The average failure scale S is expressed as Equation (16).

$$S = \frac{N_f}{N} \tag{16}$$

In Equation (16), N_f is the number of abnormal port nodes in the container shipping network during the cascade failure after failure, N is the number of port nodes in the initial network. The vulnerability of the container transport network is positively correlated with R, which satisfies $R \in [0,1]$. The higher the R, the stronger the vulnerability of the container shipping network and the lower the resistance to destruction.

Average network efficiency E is an essential indicator of the extent to which container networks are compromised. For the entire network, the average efficiency between all port nodes is the network efficiency, and the reciprocal of the shortest distance between port node i and the port node in the network is used to represent the efficiency between two points. E is calculated as shown in Equation (17).

$$E = \frac{1}{N(N-1)}\sum_{i \neq j}\frac{1}{d_{ij}} \tag{17}$$

Figure 6 shows the specific simulation process of the cascade failure effect of the container shipping network in this paper.

Step 1: Build a container shipping network and initialize the port node status of the shipping network. All port nodes are usually carried. The initial cargo volume of port node i is $Li(0)$, the capacity of port node i is $Ci(0)$, and $Li(0) < Ci(0)$.

Step 2: Select a port node in the shipping sub-network of the container shipping network. Delete the port node from the container shipping network and transfer its status to the failed port node.

Step 3: Load redistribution. According to the load redistribution principle in the cascaded failure model of a container shipping network, the interlayer transmission between the shipping network layer and the collection and distribution network layer is preferred for load redistribution. If the capacity of port nodes in the container network layer exceeds their limit capacity range, the load is redistributed to the maritime network layer with the probability of λ.

Step 4: Determine the status of port nodes. Calculate the container load of the remaining port nodes in the container shipping network. If the container load is $L_i > S_i$, the port

node is transferred to the failed port node and jumps to the third step. Supposing the container load of the port node is $C_i \le L_i < S_i$.

Step 5: The port nodes operating generally in the container shipping network are affected by the adjacent overloaded port nodes with the probability of λ.

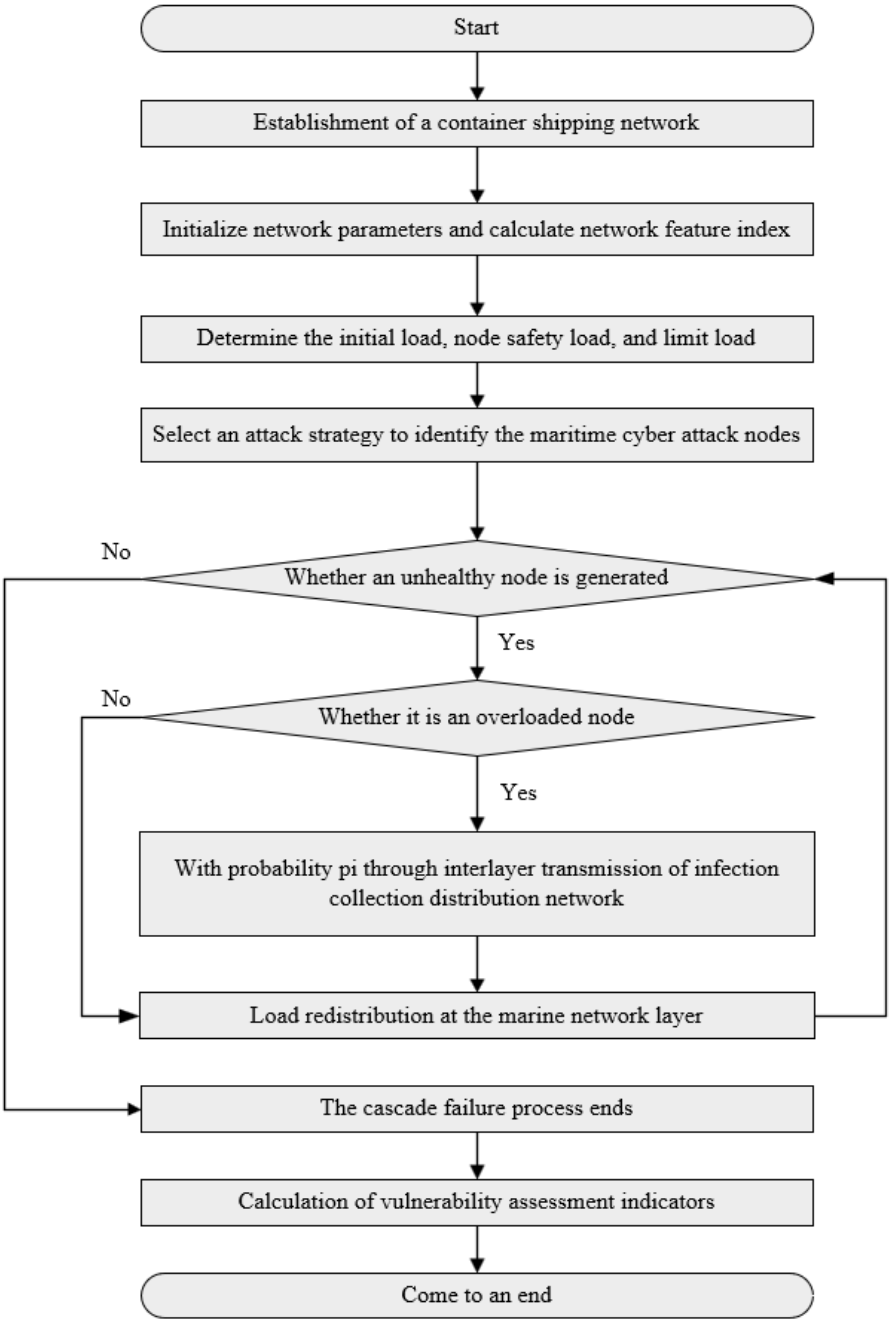


Figure 6. Simulation flow chart.

4. Empirical Results and Analysis

The incidents affecting the container shipping networks in China and the United States are categorized into two types, random attacks and deliberate attacks. Random attacks refer to a port node that is randomly selected within the network. Without a clear target in advance, a random attack is uncertain and uncontrollable. Here, the global COVID-19 pandemic outbreak event in early 2020 belongs to a random attack. A deliberate attack is an intentional act aimed at disrupting the structure and function of a container

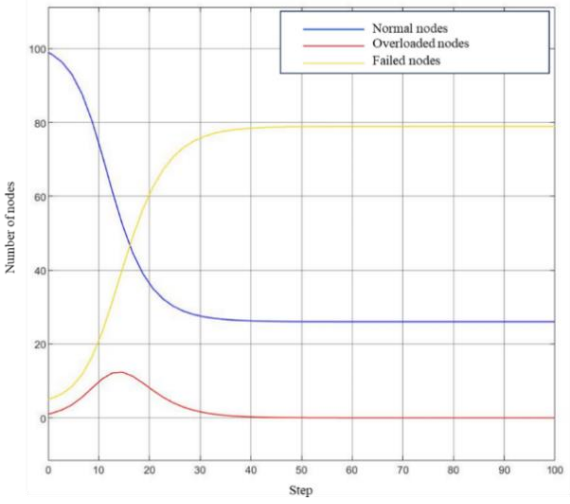
shipping network to fulfill personal goals or malicious purposes. Such attacks are usually targeted and pose a serious threat to the security and stability of the network, necessitating effective security measures to mitigate them. The Suez Canal congestion incident in March 2021 and the three major U.S. port strikes in July 2022 could be categorized as deliberate attacks occurrences. For empirical study, these three serious port congestion incidents are selected to comprehensively analyze the impacts of the two types of attacks.

4.1. Analysis of the Vulnerability Evolution of the China–U.S. Container Shipping Network

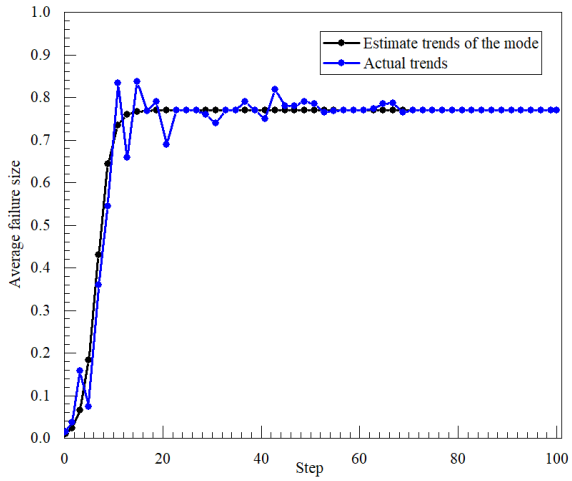
1. Global outbreaks in early 2020

In the early days of the epidemic, most shipping companies had pessimistic expectations about the international shipping situation and reduced shipping capacity, not only reducing the voyage of export containers but also significantly dismantling idle container ships. A total of 11 of the world's top 12 container shipping companies cut capacity and reduced their fleets. Many small and medium-sized shipping companies closed down one after another because they could not withstand the economic pressure caused by the long-term suspension. Later, with the spread of the epidemic worldwide, port congestion, fewer large ports ensuring the smooth flow of goods, and the problem of the slow return of empty containers became prominent. Around November 2020, the Los Angeles and Long Beach ports alone had tens of thousands of containers stranded at the docks, and port freight was "close to complete paralysis". Not only the United States, but also Singapore, Australia, and the United Kingdom had a large backlog of empty containers.

Figure 7a illustrates the estimated ratio of the density of each node during the global epidemic outbreak, comparing the model predictions with the actual changes. It reveals that the number of normal nodes, overloaded nodes, and failed nodes within the China–U.S. container shipping network fluctuates over time. Initially, upon the outbreak's impact, the number of normal nodes declined sharply, while both overloaded and failed nodes increased significantly. Subsequently, the count of overloaded nodes rose before eventually diminishing, leading the China–U.S. container shipping network to attain a stable state. Figure 7b shows a significant increase in the average failure size of the network. Upon achieving a stable state, the average size was diminished by 78% relative to the prior period, indicating that the public health crisis substantially affected the container networks of China and the United States.



(**a**)



(**b**)

Figure 7. Density of the global outbreak event at various points: (**a**) The estimated ratio of density at each node under the global outbreak event. (**b**) The comparison between the estimated trend of the mode and the actual trend.

Following the outbreak of the epidemic, the abrupt and extensive failure of ports led to a rapid decline in the number of normal nodes within the container shipping network. The proliferation of overloaded and failed nodes ensued, and the blockade measures due to the epidemic resulted in significant operational disruptions at numerous ports, directly impacting the loading, unloading, and transportation of goods. Port operational systems were impacted by human management and traffic limitations, resulting in decreased cargo transport efficiency. Consequently, once China and the United States initiated their emergency responses, the container shipping network between the two nations swiftly returned to a condition of balance. The management of staff and the requisite protective measures instituted due to the outbreak resulted in diminished port capacity, an inadequate port yard consolidation system, and a decline in the efficiency of maritime operations.

2. The Suez Canal congestion event in March 2021

As one of the busiest canals in the world, about 12% of global trade, 15% of cargo ships shipped by sea, and about 30% of seaborne oil pass through the Suez Canal. Not long after the last "Long" blockage, an Italian oil tanker became stuck in the canal because of an engine failure, and the Suez Canal faced another congestion. In the short term, regarding the subsequent impact of the Suez Canal blockage, the relevant shipping companies believe that the effect on Europe, the Mediterranean, and the eastern route of the United States will take 6 to 7 days, and the effect on the shipping schedule maybe 1 to 2 weeks, and the subsequent backlog of goods, a large number of ships, and congestion may last for more than a month.

Figure 8a illustrates the estimated density ratio of each node during the Suez Canal congestion event based on a comparison between the model predictions and the actual changes. Initially, in the China–U.S. container shipping network, the congestion event led to a gradual decline in the number of normal nodes, while the increase in overload and failure nodes also occurred at a slow rate. The failure nodes increased at a gradual pace, indicating that the container shipping network was not significantly impacted by the early congestion in the Suez Canal. The network experienced a diverse range of port node anomalies due to the epidemic, yet its operation remained relatively normal without widespread port node failures. However, as the congestion in the Suez Canal persisted over time, the risk of diffusion within the container shipping network escalated, resulting in a higher degree of network failure. Ultimately, as time progressed, the network stabilized. Figure 8b illustrates that the average magnitude of network failure increased gradually, and upon the re-establishment of stability, the

average network size diminished by 59% relative to the preceding period, indicating that prolonged ship stranding significantly affected container shipping.

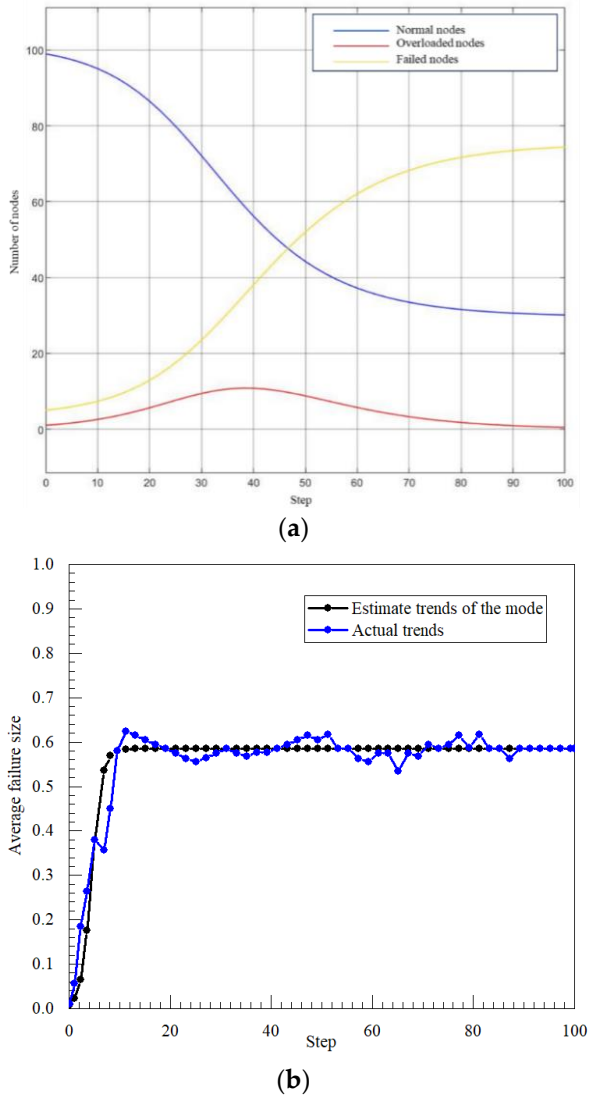


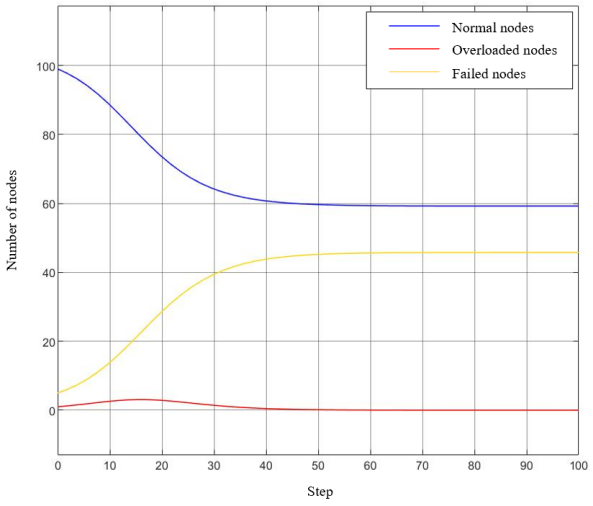
Figure 8. Density at each node of the Suez Canal congestion event: (**a**) The estimated ratio of density at each node under the Suez Canal congestion event. (**b**) The comparison between the estimated trend of the mode and the actual trend.

During the Suez Canal congestion event, the number of normal nodes in the network gradually diminished in the initial phase, while the count of overloaded and failed nodes climbed at a more gradual pace. Nevertheless, as the danger of failure proliferated throughout the network, a greater number of nodes became impacted. The Suez Canal offers the most direct passage from Europe to the Indian Ocean and adjacent territories in the Western Pacific Ocean, minimally affecting the China–U.S. container routes geographically. However, due to the proliferation of risks within the global container shipping network, the China–U.S. container shipping system experienced irregularities at ports. The prolonged stranding of vessels, particularly in container transport, significantly impacts operations. The stranding resulting from Suez Canal congestion will inevitably disrupt the shipping schedules of certain companies. The necessary detours will extend voyage duration and incur elevated fuel expenses.

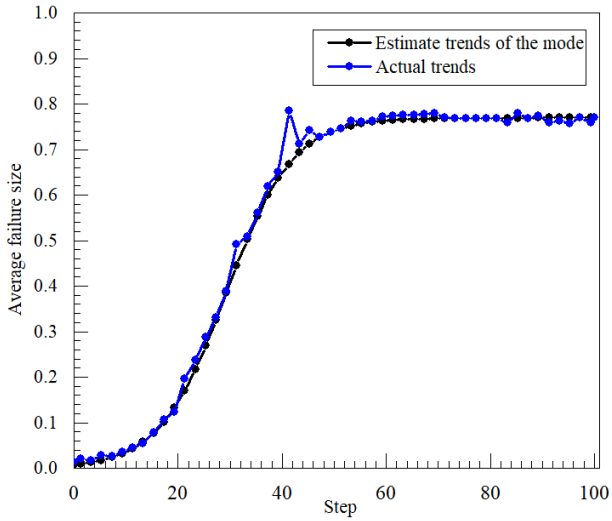
3. The strike at three major U.S. ports in July 2022

As negotiations between the International Longshoremen's Association (ILA) and the United States Maritime Union (USMX) continue to deadlock over wage increases and port automation, a severe strike at U.S. Eastern and Gulf Coast ports seems inevitable. Linerlytica, a shipping analyst, pointed out that given the current situation, a strike is almost inevitable, highlighting that the 14 ports controlled by the ILA handled 28.4 million TEUs of containerized cargo in 2023, with a weekly throughput of nearly 550,000 TEUs. Each extended week of the strike will halt about 1.7 percent of the global container fleet.

Figure 9a shows the comparison between the model predictions and the actual changes in the three major U.S. port strike events, indicating that the China–U.S. container shipping network experienced minimal disruption in the pre-strike period of these ports. Over time, the quantities of normal nodes, overloaded nodes, and failed nodes within the China–U.S. container shipping network fluctuate. During the initial phase of the strike events at the three major U.S. ports, the decline in normal nodes across the entire network was gradual, while the increases in overloaded and failed nodes were also slow. This indicates that the container shipping network between China and the U.S. has not been significantly affected by the strikes at U.S. ports. Nonetheless, as the port strike progresses into its later stages, its impacts on the container shipping network are becoming increasingly apparent. This event exerts a persistent influence on the container shipping network, with the restoration to a stable state occurring at a slower pace than that observed during the pandemic. Figure 9b illustrates that the average failure size of the network increases gradually, and upon achieving a stable state, the average size of the network diminishes by 79% relative to the preceding period. This indicates that the port strike event exhibits a degree of persistence.



(**a**)



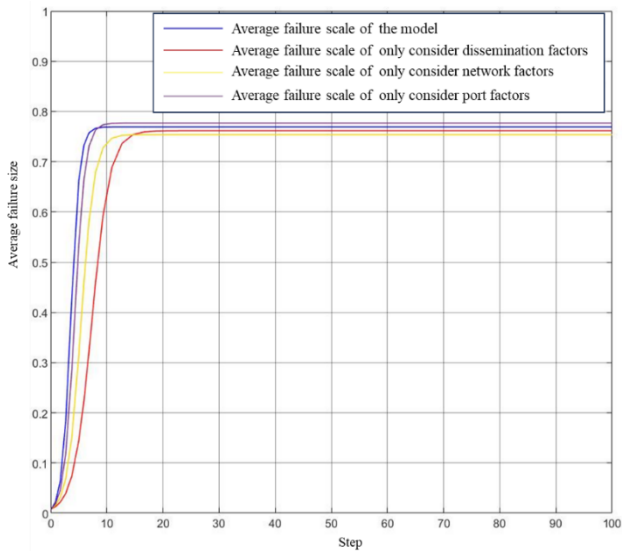
(**b**)

Figure 9. Density at each point of the strike at the three major U.S. ports: (**a**) The estimated ratio of density at each point under the strike at the three major U.S. ports. (**b**) The comparison between the estimated trend of the mode and the actual trend.

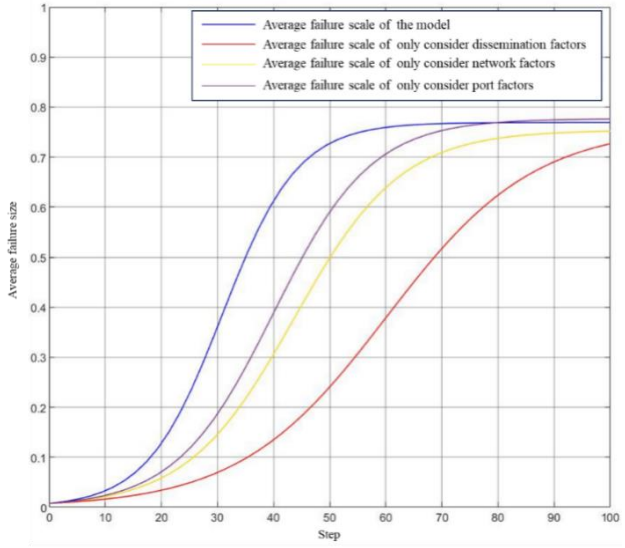
The strike was a strategy employed by port workers seeking to preserve their status as independent contractors. The strike was brief, however, due to the substantial volume of shipments, and the accumulation of cargo at the terminals, the China–U.S. container shipping network experienced great disruption, resulting in additional irregularities. The combined effects of a strike and an outbreak may diminish port collection and evacuation capacity, as well as the effectiveness of ship operations, resulting in losses to the shipping network. The optimization of the port collecting and distributing system is crucial for enhancing operational efficiency and minimizing logistical expenses. Furthermore, port capacity directly influences the stability and resilience of the overall maritime network against damage. The strike has diminished the port's capacity, disrupted the collection and distribution system inside the port area, and decreased the efficiency of ship operations, resulting in substantial losses to the whole shipping network.

4.2. Impact of Operational Risk Propagation Probability on the Vulnerability of the Container Shipping Network

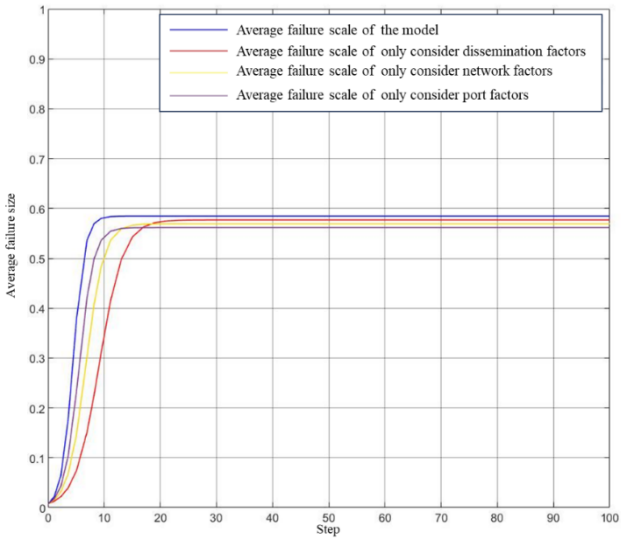
To measure the influence of port, ship, and network factors on the diffusion and propagation of congestion risk, each element is used to experiment one by one to explore the influence of different factors on the dynamic change trend of overloaded ports. As shown in Figure 10, the abscissa represents the time series, and the ordinate represents the average failure size of the container shipping network.



(**a**)



(**b**)



(**c**)

Figure 10. The three events influenced trend changes in each factor: (**a**) Global outbreak events at various points. (**b**) The Suez Canal congestion event. (**c**) The strike at the three major U.S. ports.

When any factors influencing operational risk diffusion are considered separately, the rate at which the network reaches a new equilibrium state slows significantly. Still, it has a negligible impact on the average failure scale of the original network. The shipping network is the ship's carrier, so each vessel's congestion risk can be developed into the risk of congestion accidents on a route. The propagation characteristics of congestion risk indicate that there is a specific change law with the change in shipping traffic status. It is worth noting that congestion risk is widely spread in the network, and the average impact of the initial diffusion node is significant, so once the hub port is affected by congestion, it has a much more substantial effect on the entire shipping network than the non-hub port affected by congestion. When only port factors are considered, the impact of port cargo volume on the vulnerability of the container shipping network is mainly considered in this paper. Compared with the port's cargo volume, the ship's service level has a more noticeable effect on the ship at risk of congestion. Therefore, it is necessary to improve the level of ship service capacity to reduce the vulnerability of the container shipping network, especially the port punctuality rate. A ship that does not call at the port on time will produce a domino effect, and the overall ship turnover rate will be significantly reduced, and finally, the total market capacity will be reduced.

When port nodes become overloaded, the vulnerability of the container shipping network is stimulated, and then the efficiency is reduced, and the risk increases. Due to the fixed route of the container shipping network, this will significantly impact the upstream and downstream port nodes.

The operational risk propagation probability λ acts on the expected port node, which increases the decline risk of the port's functional efficiency, affecting the port node's regular operation, and expanding the influence scope of cascade faults in the container shipping network. Under different operational risk propagation probabilities, the cascade failure vulnerability of the container shipping network is simulated and analyzed, and the limit coefficient of port nodes $\alpha_2 = 1$ is obtained. Taking the global public health emergency in early 2020 as an example in the simulation study. It was found that the average failure scale of the container shipping network is relatively large when attacked by port node degree and the intermediate number of port nodes. This method is not conducive to spreading operational risks, so the random attack method is chosen to simulate the container shipping network.

The influence trend of operational risk propagation probability λ on the average failure scale of the network is stable in Figure 11. With the increase in λ, the impact of cascade failure effect on the container shipping network in China and the United States and the relative size of the average failure scale of the container shipping network in China and the United States both increases. This shows that in the cascading failure effect of the container shipping network, the vulnerability of the container shipping network between China and the United States is stimulated due to overloaded port nodes and failed port nodes, resulting in reduced network turnover efficiency and increased operational risks. In the container shipping network between China and the United States, the substantial coupling degree between ports will inevitably affect adjacent port nodes and expand the influence scope of the cascade failure effect of the network. If λ is high ($\lambda > 0.35$), the increase in the tolerance coefficient α_1 cannot effectively reduce both the average failure scale of the container shipping network and the vulnerability of the container shipping network. This dovetails with the theory of operational risk propagation thresholds. When λ exceeds its threshold, the port operation impact caused by overloaded port nodes will spread rapidly in the container shipping network, expanding the scope of the cascade failure effect. When $\lambda > 0.5$, no matter how significant the tolerance coefficient is, it will cause a rapid collapse of the network. When many ports collapse or an extensive affected range occurs, the propagation of operational risks can quickly occur throughout the network, resulting in a rapid collapse of the network.

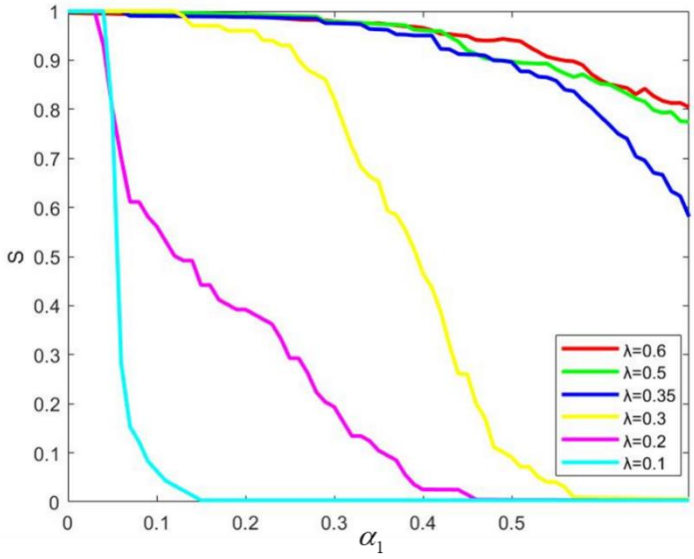


Figure 11. Impact of infection probability on network vulnerability.

4.3. Impact of Tolerance and Limit Coefficients on the Vulnerability of the Container Shipping Network

First, the impact of the tolerance coefficient α_1 of port node capacity on risk propagation is analyzed when the cascade failure of the container shipping network occurs in China and the United States. According to the simulation results, the vulnerability of the network is analyzed, and the simulation results are shown in Figure 12.

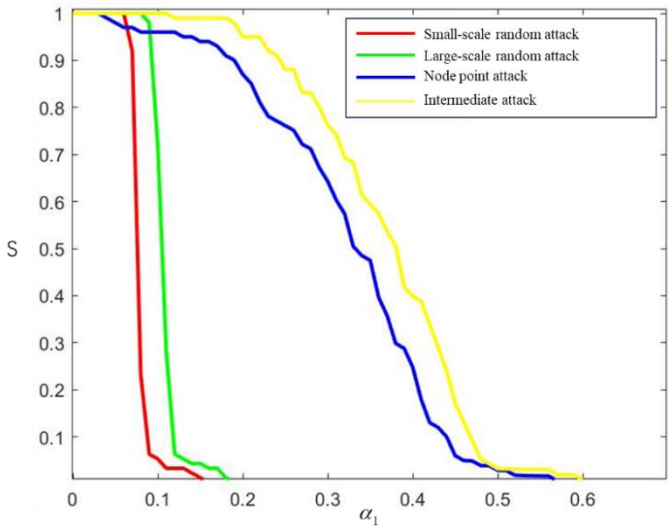


Figure 12. Impact of tolerance factor on network vulnerability.

If the tolerance factor of the port node is small, the average failure scale of the container shipping network is higher. When the tolerance coefficient of the port node $\alpha_1 = 0.05$, the average failure scale of the container shipping network in China and the United States is 100% in the four attack modes, and the container shipping network can be determined to be in a collapsing state. With the increase in α_1, the average failure scale of the container shipping network is negatively correlated. That is, the vulnerability of the overall container shipping network decreases. When α_1 is large enough, that is, when the capacity of the port yard is large enough, such as $\alpha_1 = 0.6$, no matter which attack method is used against the China–U.S. container shipping network, the China–U.S.

container shipping network does not have a cascading failure effect. The average failure scale of the network has not changed, the network stability is currently solid, and the vulnerability is extremely low. In reality, considering the cost of port construction, during the period of port planning and construction, the cascade failure effect can be controlled within a specific range by improving the tolerance coefficient of port nodes. By doing this, the port and shipping logistics system can be prevented from being paralyzed due to the collapse of a large number of terminals.

Meanwhile, Figure 12 shows that whether it is a large-scale random attack or a small-scale random attack, compared with the unexpected attack method, the average failure scale of the container shipping network in China and the United States is a little larger; that is, the vulnerability is more substantial. For example, when $\alpha_1 = 0.35$, the average failure size of the container shipping logistics network is 0.61 under the maximum intermediate attack mode, while the average failure size of the container shipping network is 0 under random attack mode. This indicates that there are several port nodes with low throughput and low importance in the container shipping network between China and the United States. Due to random attacks, container loads of these port nodes are redistributed to their neighboring nodes. The additional load pressure of these ports on other ports is slight, so there is no cascading failure effect. Therefore, such port nodes with low throughput and low importance are less affected by the risk propagation of cascaded failure operations.

Secondly, the impact of the ultimate carrying capacity of port nodes in the container shipping network in China and the United States on the vulnerability of the container shipping network in China and the United States is discussed. In the experiment, the limit coefficient of the port node α_2 is analyzed, then the tolerance coefficient $\alpha_1 = 0.35$ and the probability of operational risk propagation $\lambda = 0.3$ are obtained. Moreover, the impact of different limit coefficients on the vulnerability of the container shipping network in China and the United States is analyzed under the two attack modes of random attack and deliberate attack, where the node attack method in a deliberate attack is selected in this paper. The simulation results are shown in Figures 13 and 14.

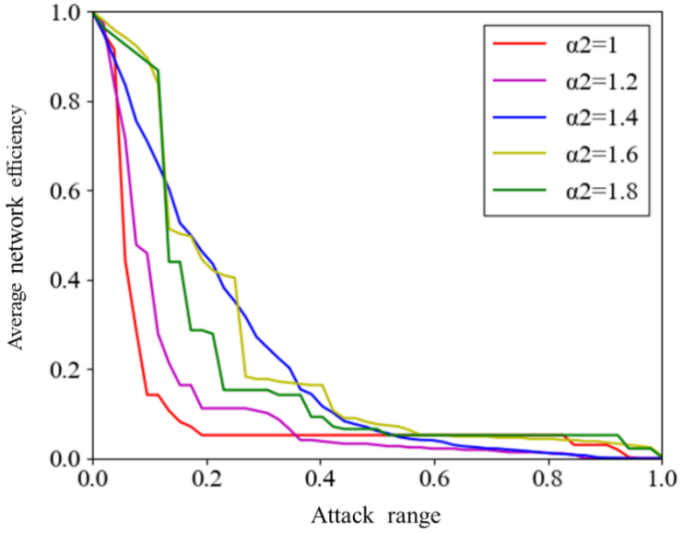


Figure 13. Impact of different limit coefficients on the vulnerability of the China–U.S. container shipping network under random attacks.

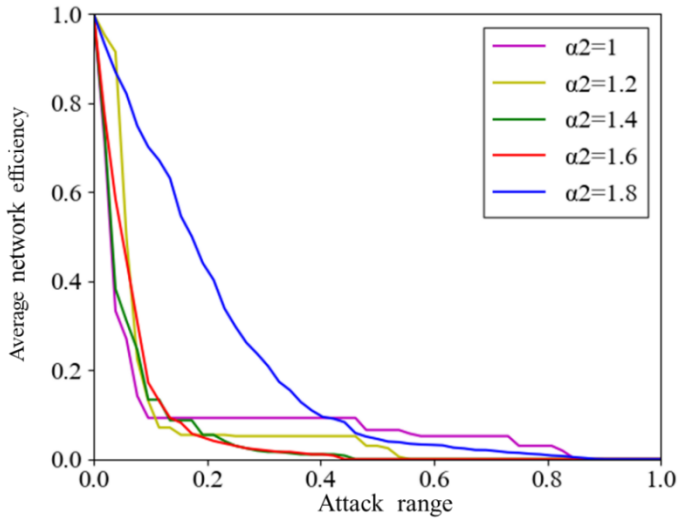


Figure 14. Impact of different limit factors on the vulnerability of the China–U.S. Container shipping network under deliberate attacks.

The above two figures show that the limit coefficient of the port node α_2 and the simulation result trend of the tolerance coefficient α_1 are basically the same. When the limit coefficient of the China–U.S. container shipping network α_2 increases, the average failure scale of the network decreases, which indicates that the invulnerability of the network is gradually improving and the vulnerability is slowly reducing. Through the case analysis of the random attack mode in Figure 14, when $\alpha_2 = 1$, 10% of the port nodes are removed in the container shipping network of China and the United States, and this will cause the average network efficiency of the container shipping network of China and the United States to drop below 0.2. When $\alpha_2 = 1.8$, 30% of port nodes in the network are removed, and it will cause the average network efficiency of the container shipping network of China and the United States to drop below 0.2. This shows that when α_2 increases, the scope of the container shipping network in China and the United States affected by the cascading failure effect gradually decreases, and the vulnerability of the container shipping network in China and the United States is also lower, making the network more stable. Due to the increase in α_2, the average network efficiency of the container shipping network slows down. This indicates that as the increase in the value of allowing port nodes to exceed their operating volume, the impact of random attacks on the operational risk propagation of the container shipping network in China and the United States decreases, and the invulnerability of the container shipping network in China and the United States increases.

5. Conclusions

Considering the uncertainty, complexity, and destructiveness of emergencies, a double-layer coupled container network is created. Then, this paper constructs a cascading failure model of a container shipping network considering risk level to characterize the impact transmission mechanism of emergencies on the container shipping network. The main conclusions of this paper are as follows.

According to the characteristics of the container shipping network, a load redistribution method for failed nodes of the container shipping network is proposed based on the load capacity model and infectious disease model. After introducing this method into the cascade failure model of the container shipping network, it is found that when the throughput of the port node is less than its safe capacity, the port node will not fail. When the throughput of a port node exceeds its safe capacity but is less than the limit capacity,

the port node fails with a certain probability. When the throughput of a port node exceeds its limit capacity, the port node immediately fails.

As an example, this paper selects three serious port congestion events in the container shipping network between China and the United States. It compares the trend of actual data with the model's predicted direction. The results show that the expected movement of the cascade failure model of the container shipping network, considering the risk level, is consistent with the actual trend. This paper explains the container shipping network model based on the probability of operational risk propagation, which aligns with reality. The tolerance and limit coefficients of port nodes have a significant negative correlation with the average failure scale of the container shipping network. When these two parameters increase, the average failure size of the container shipping network becomes smaller, which means that the vulnerability of the container shipping network becomes lower, and the cascade failure effect affects the container shipping network less. To reduce the exposure of the container shipping network, the capacity of ports should be appropriately increased, and the limit capacity of facilities and equipment in existing ports should be improved. When random attacks occur on the container shipping network between China and the United States, the container shipping network between China and the United States has anti-destruction solid ability and low vulnerability. When port node degree attacks or intermediate attacks occur, the container shipping network has weaker anti-destruction ability and more robust exposure than random attacks. In addition, as the probability of port operation risk transmission increases, the cascading failure effect of the container shipping network is more affected. When the probability of port operation risk propagation exceeds its threshold, even if the capacity of port nodes is increased, the average failure scale of the container shipping network cannot be reduced. The vulnerability of the container shipping network cannot be improved, which proves the correctness of the definition of the three states of port nodes in this paper.

With the deepening of container shipping network operation, multi-agents in the shipping network usually adopt shipping alliances to improve competitiveness. In addition, the relationships between ports and ports, ports and routes, routes and routes are becoming more and more complex. In this paper, when establishing the cascading failure model of a container shipping network considering risk level, only the influence of port container throughput redistribution on port node state is considered, and the complexity of the interaction relationship between participants and nodes in the container shipping network is not considered. It is also necessary to conduct more in-depth research on the competition and cooperation relationship between port nodes and the mutual restriction relationship between ports and routes, so as to analyze the change rule of node state.

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